

Yuxiang Mao

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🎓 Education

Northeastern University

Sep 2023 – Jul 2027 (expected)

School of Computer Science and Engineering

Computer Science and Technology

B.S. Candidate

💼 Work Experience

ByteDance Agent Development Intern

Apr 2026 – Present

Global Monetization · Platform Quality · Creative and Ecosystem

- Built backend services for a front-end self-testing agent, delivering automated testing to business teams via the A2A (Agent-to-Agent) protocol across case generation, execution, and assertion.
- Owned context engineering for the evaluation agent, architecting context-window management and knowledge recall to keep task information and execution context intact across long-horizon runs.
- Implemented the core Case Run skill with multi-agent orchestration, introducing Retry Round retries that raised evaluation case success rate from 10% to 60%.

🔧 Projects

TaskWizard Android GUI Agent Framework

Jun 2026 – Present

Personal open-source project Python / LangChain / ADB / MLX

[GitHub: TaskWizard](#)

- **Architecture Refactor:** Evolved Open-AutoGLM into an LLM-driven thin loop, replacing fixed node orchestration with model-selected tools while the runtime owns execution, context, and risk alerts; on-device success on 15+ minute tasks rose from 20% to 90%.
- **Pluggable Runtime:** Composed safety, budget, and memory modules through a capability registry and event bus, enabling tool swaps, model Provider/API Builder extensions, and per-role model configuration for execution, summarization, and acceptance.
- **Multimodal Grounding:** Fused screenshots with the Accessibility Tree, validating mark freshness via batch identifiers; wrapped LocateAnything as a visual grounding tool with container/anchor cropping and coordinate remapping, covering missing-widget and ambiguous-grounding cases.
- **Long-Horizon Context:** Implemented scroll-based screenshot eviction and mark folding with the TaskDoc task board pinned; two-threshold auto-compaction generates handoff summaries, preserves tool-call/receipt pairs, and accumulates per-role token usage across compactions.
- **Experience Iteration:** Built the App-KB alias library and episode archive with recall over sqlite-vec/FTS5 and local MLX embeddings; offline-distilled rules and process cards are model-reviewed, graded, and reinjected under control, with shadow evaluation, versioning, and rollback.
- **Runtime & Diagnostics:** Isolated the execution engine in a standalone runner subprocess, decoupled from the NiceGUI console via file IPC; supports task reconnection, event replay, tool-receipt diagnostics, and redacted traces for post-mortem review.

⚙️ Skills

- **Programming Languages:** Python, Go, C++
- **Agent Engineering:** LangChain/LangGraph, Tool Calling, Context Engineering, RAG/Memory Systems, Multi-Agent Collaboration, VLM/GUI Grounding
- **Backend & Tooling:** Linux, Docker, PostgreSQL, Gin, Git/GitHub Actions, ADB; hands-on React experience
- **Language Proficiency:** English (IELTS 7.0)